

CHAPTER 15

Factorial Designs

A factorial design has some number of factors occurring at some number of levels. In a full factorial design, all possible combinations of the levels of the factors occur at least once. We start with experiments involving two factors and then look at designs with more than two factors.

15.1 Two-Way ANOVA

Suppose we have two factors, α at I levels and β at J levels. Let n_{ij} be the number of observations at level i of α and level j of β and let those observations be $y_{ij1}, y_{ij2}, \dots$ etc. A complete layout has $n_{ij} \geq 1$ for all i, j . A balanced layout requires that $n_{ij} = n$.

The most general model that may be considered is:

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\beta\alpha)_{ij} + \epsilon_{ijk}$$

As in the one-way layout, not all the parameters are identifiable. The interaction effect $(\beta\alpha)_{ij}$ is interpreted as that part of the mean response not attributable to the additive effect of α_i and β_j . For example, you may enjoy strawberries and cream individually, but the combination is far superior. In contrast, you may like fish and ice cream, but not together.

If the main effects α and β generate design matrices X_α and X_β , then the design matrix for the $\beta\alpha$ interaction is given by collecting the element-wise products of all columns of X_α with all columns of X_β to form a matrix with $(I-1)(J-1)$ columns.

A significant interaction makes the model hard to interpret, as $\hat{\alpha}$ cannot be studied independent of $\hat{\beta}$. A comparison of the levels of α will depend on the level of β . Consider the following two layouts of $\hat{\mu}_{ij}$ in a 2×2 example:

	Male	Female	Male	Female
Drug 1	3	5	2	1
Drug 2	1	2	1	2

The response is a measure of performance. In the case on the left, we can say that drug 1 is better than drug 2 although the interaction means that its superiority over drug 2 depends on the gender. In the case on the right, which drug is best depends on the gender. We plot this in Figure 15.1. We see that in both cases the lines are not parallel, indicating interaction. The superiority of drug 1 is clear in the first plot and an ambiguous

conclusion is seen in the second. Make plots like this when you want to understand an interaction effect.

When the interaction is significant, the main effects cannot be defined in an obvious and universal way. For example, we could define the gender effect as the effect for females,

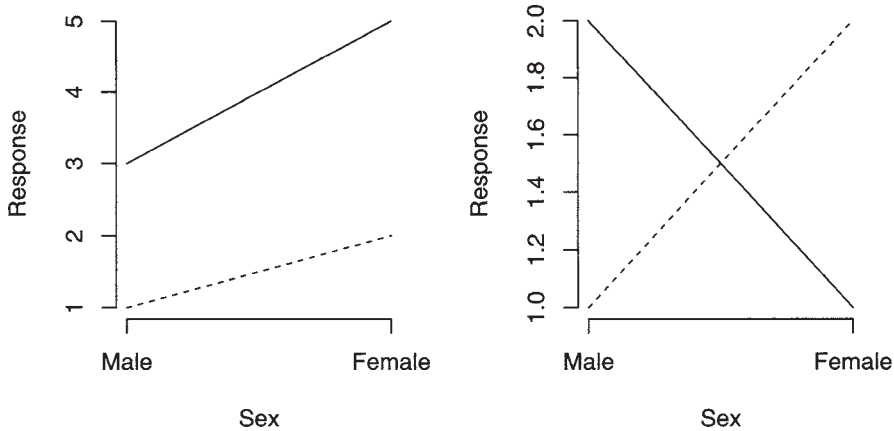


Figure 15.1 Two 2×2 tables with the response plotted by the factors, sex on the horizontal axis and drug 1 as the solid line and drug 2 as the dotted line.

the effect for males, the effect for the average of males and females or something else. If there was no interaction effect, the gender effect could be defined unambiguously.

When you have a significant interaction, an interaction plot of the fitted values can make it easier to describe the nature and impact of the effect. Alternatively, you can fit a model:

$$y_{ijk} = \mu_{ijk} + \varepsilon_{ijk}$$

and then treat the data as a one-way ANOVA with IJ levels. Obviously this makes for more complex comparisons, but this is unavoidable when interactions exist.

15.2 Two-Way ANOVA with One Observation per Cell

Mazumdar and Hoa (1995) report an experiment to test the strength of a thermoplastic composite depending on the power of a laser and the speed of a tape:

```
> data(composite)
> summary(composite)
  strength laser  tape
Min.   :20.6 40W:3 slow :3
1st Qu.:28.0 50W:3 medium:3
Median :29.8 60W:3 fast :3
Mean    :31.0
```

3rd Qu.:35.7

Max. :39.6

We can fit a model with just the main effects as:

```
> g <- lm (strength ~ laser + tape, composite)
```

However, if we tried to add an interaction term, we would have as many observations as parameters. The parameters could be estimated, but no further inference would be possible. Nevertheless, the possibility of an important interaction exists. There are two ways we can check for this.

The interaction can be checked graphically using an *interaction plot*. We plot the cell means on the vertical axis and the factor α on the horizontal. We join the points with same level of β . The role of α and β can be reversed. Parallel lines on the plot are a sign of a lack of interaction as seen in the plots in Figure 15.2:

```
> with (composite, interaction.plot (laser, tape, strength, legend=F))
> with (composite, interaction.plot (tape, laser, strength, legend=F))
```

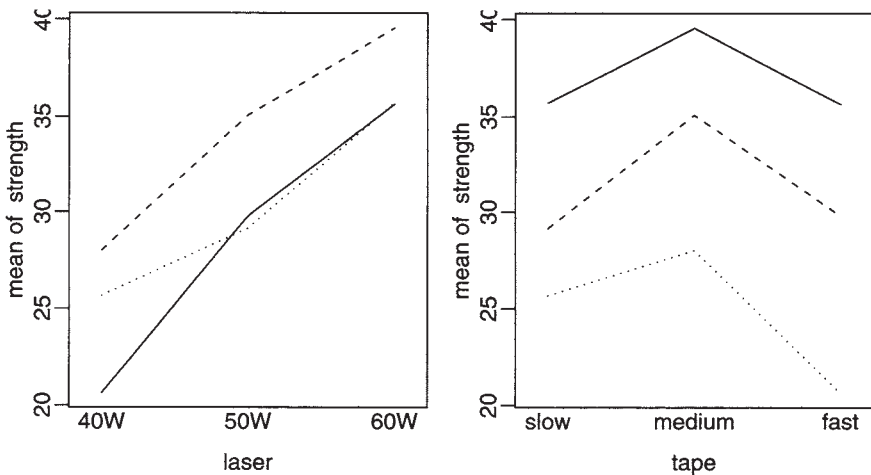


Figure 15.2 *Interaction plots for the composite data.*

Because of random variation, we cannot expect the lines to be exactly parallel. Allowing for a certain amount of noise, these lines are approximately parallel, so we conclude there is no substantial interaction between the factors.

Tukey's nonadditivity test provides another way of investigating an interaction. The model:

$$y_{ij} = \mu + \alpha_i + \beta_j + \phi_i \beta_j + \varepsilon_{ijk}$$

is fit to the data and then we test if $\phi = 0$. This is a nonlinear model because it involves the product of parameters, $\phi_i \beta_j$. Furthermore, it makes the assumption that the interaction effect is multiplicative in form. We have no particular reason to believe it takes this form and so the alternative hypothesis may not be appropriate. We execute the test:

```
> coef (g)
(Intercept) laser50W laser60W tapemedium tapefast
 23.9178      6.5733  12.2133      4.0367 -1.4800
> lasercoefs <- rep(c(0,6.5733, 12.2133),3)
> tapecoefs <- rep(c(0,4.0367,-1.4800), each=3)
> h <- update (g, . ~ . + I (lasercoefs*tapecoefs)
> anova (h)
Analysis of Variance Table
Response: strength
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
laser	2	224.2	112.1	36.82	0.0077
tape	2	48.9	24.5	8.03	0.0624
I (lasercoefs * tapecoefs)	1	1.4	1.4	0.45	0.5503
Residuals	3	9.1	3.0		

The p-value of 0.55 indicates a nonsignificant interaction. So for these data, we can reasonably assume $(\beta_{ij}) = 0$. We can now test the main effects:

```
> anova (g)
Analysis of Variance Table
Response: strength
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
laser	2	224.2	112.1	42.69	0.002
tape	2	48.9	24.5	9.32	0.031
Residuals	4	10.5	2.6		

We see that both factors are significant. Examining the coefficients above, we see that the strength increases with the laser power but the strength is largest for medium tape speed but less for slow or fast tape speeds.

The treatment coding does not take advantage of the ordered nature of both factors. We can declare both to be *ordered factors* and refit:

```
> coraposite$laser <- as.ordered(composite$laser)
> composite$tape <- as.ordered(composite$tape)
> g <- lm (strength ~ laser + tape, composite)
> summary (g)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    31.032      0.540    57.45 5.5e-07
laser.L         8.636      0.936     9.23 0.00077
laser.Q        -0.381      0.936    -0.41 0.70466
tape.L         -1.047      0.936    -1.12 0.32594
tape.Q         -3.900      0.936    -4.17 0.01404
Residual standard error: 1.62 on 4 degrees of freedom
```

Multiple R-Squared: 0.963, Adjusted R-squared: 0.926
 F-statistic: 26 on 4 and 4 DF, p-value: 0.00401

Instead of a coding with respect to a reference level, we have linear and quadratic terms for each factor. The coding is:

```
> contr.poly (3)
      .L      .Q
[1,] -7.0711e-01  0.40825
[2,] -9.0733e-17 -0.81650
[3,]  7.0711e-01  0.40825
```

We see the linear term is proportional to $(-1, 0, 1)$ representing a linear trend across the levels while the quadratic term is proportional to $(1, -2, 1)$ representing a quadratic trend.

We see that the quadratic term for laser power is not significant while there is a quadratic effect for tape speed. One of the drawbacks of a model with factors is the difficulty of extrapolating to new conditions. The information gained from the ordered factors suggests a model with numerical predictors corresponding to the level values:

```
> composite$Ntape <- rep (c (6.42,13,27), each=3)
> composite$Nlaser <- rep (c(40, 50, 60), 3)
> gl <- lm (strength ~ Nlaser + poly(log(Ntape),2), composite)
> summary (gl)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      0.4989      3.0592   0.16  0.87684
Nlaser            0.6107      0.0604  10.11  0.00016
polydog (Ntape),2)1 -1.8814      1.4791  -1.27  0.25933
polydog (Ntape),2)2 -6.7364      1.4791  -4.55  0.00609
Residual standard error: 1.48 on 5 degrees of freedom
Multiple R-Squared: 0.961, Adjusted R-squared: 0.938
F-statistic: 41.5 on 3 and 5 DF, p-value: 0.000587
```

We use the log of tape speed, as this results in roughly evenly spaced levels. This model fits about as well as the two-factor model but has the advantage that we make predictions for values of tape speed and laser power that were not used in the experiment. The earlier analysis with just factors helped us discover this model, which we may not otherwise have found.

15.3 Two-Way ANOVA with More than One Observation per Cell

Consider the case when the number of observations per cell is the same and greater than one, so that $n_{ij} = n > 1$ for all i, j . Such a design is orthogonal. Orthogonality can also occur if the row/column cell numbers are proportional.

With more than one observation per cell, we are now free to fit and test the model:

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\beta\alpha)_{ij} + \varepsilon_{ijk}$$

The interaction effect may be tested by comparison to the model:

$$y_{ijk} = \mu + \alpha_i + \beta_j + \varepsilon_{ijk}$$

and computing the usual F-test. If the interaction effect is found to be significant, do not test the main effects even if they appear not to be significant. The estimation of the main effects and their significance is coding dependent when interactions are included in the model.

If the interaction effect is found to be insignificant, then test the main effects, but use RSS/df from the full model in the denominator of the F-tests—this has been shown to maintain the type I error better. So the F-statistic used is:

$$F = \frac{(RSS_{small} - RSS_{large}) / (df_{small} - df_{large})}{\hat{\sigma}_{full}^2}$$

In an experiment to study factors affecting the production of the plastic polyvinyl chloride (PVC), three operators used eight different devices called resin railcars to produce PVC. For each of the 24 combinations, two samples were produced. The response is the particle size of the product. The experiment is described in Morris and Watson (1998).

We make some plots, as seen in Figure 15.3:

```
> data (pvc)
> attach (pvc)
> stripchart (psize ~ resin, xlab="Particle size", ylab="Resin railcar")
> stripchart (psize ~ operator, xlab="Particle size", ylab="Operator")
```

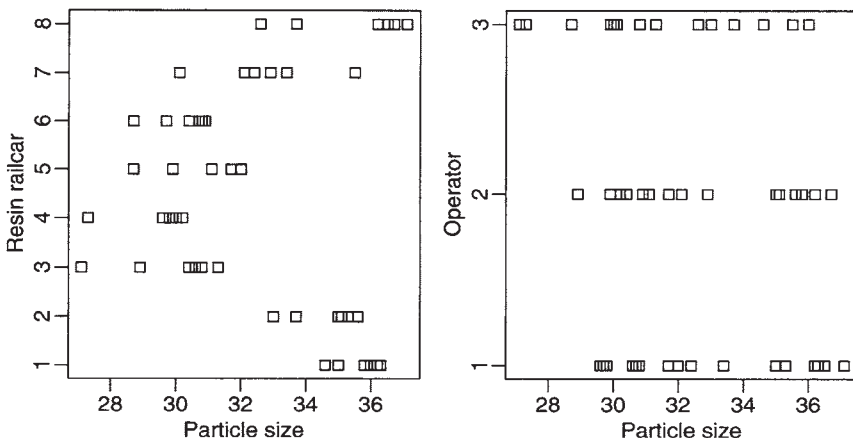


Figure 15.3 *PVC data plotted by resin railcars on the left and by operators on the right.*

Is there an interaction between operators and resin railcars? We first look at this graphically with interaction plots, as seen in Figure 15.4:

```
> interaction.plot (operator, resin, psize)
> interaction.plot (resin, operator, psize)
```

These are approximately parallel. The trouble with interaction plots is that we always expect there to be some random variation so it is sometimes difficult to distinguish true interaction from just noise. Fortunately, in this case, we have replication so we can directly test for an interaction effect.

Now fit the full model and see the significance of the factors:

```
g <- lm (psize operator*resin)
anova (g)
```

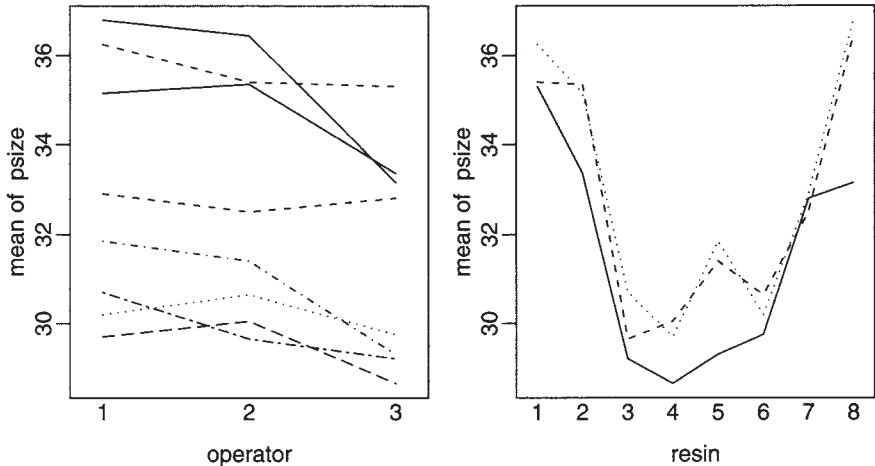


Figure 15.4 *Interaction plots for the PVC data.*

Analysis of Variance Table

Response: psize

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
operator	2	20.7	10.4	7.01	0.004
resin	7	283.9	40.6	27.44	5.7e-10
operator:resin	14	14.3	1.0	0.69	0.760
Residuals	24	35.5	1.5		

We see that the interaction effect is not significant, but the main effects are. If you remove the interaction and then retest the main effects, you get a somewhat different result. As discussed earlier, the former test is preferred (although in this particular example, it does not affect the conclusion):

```
> anova(lm(psize ~ operator + resin, pvc))
Analysis of Variance Table
Response: psize
      Df Sum Sq Mean Sq F value Pr(>F)
operator  2   20.7    10.4     7.9 0.0014
resin     7  283.9    40.6    30.9 8.1e-14
Residuals 38   49.8     1.3
```

We check the diagnostics, as seen in Figure 15.5:

```
> qqnorm (residuals (g))
> qqline (residuals (g))
> plot (fitted (g), residuals (g), xlab="FittedII",
ylab="Residuals")
```

We see some evidence of outliers. The symmetry in the residuals vs. fitted plot is because, for each combination of the factors, the mean of the two replicates is the fitted value. The two residuals for that cell will be mirror images. If we exclude the two largest

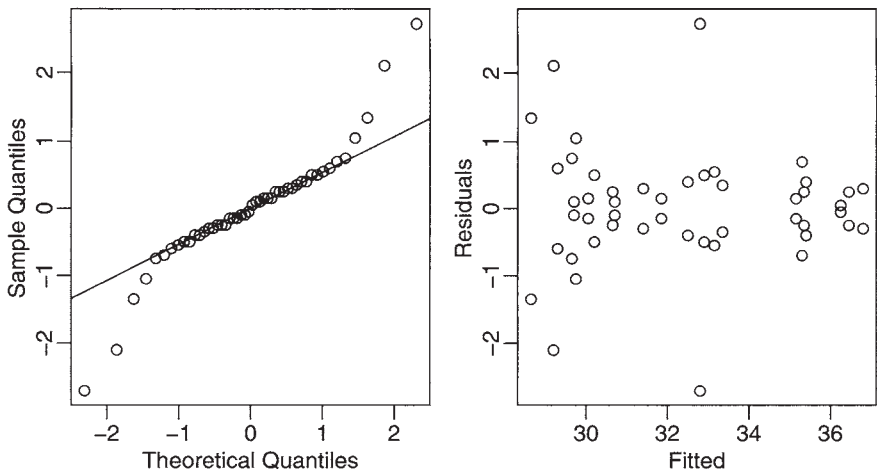


Figure 15.5 *Diagnostics plots for the full model for the PVC data.*

outliers, then the interaction is still insignificant. An examination of the main effects model reveals that case 45 is the outlier, which we now exclude:

```
> g <- lm (psize ~ operator+resin,subset=-45)
> summary (g)
Coefficients:
      Estimate Std. Error t value Pr(>|t|)
(Intercept)  36.337      0.430   84.60 < 2e-16
operator2    -0.263      0.332   -0.79  0.435
operator3    -1.797      0.339   -5.30 5.5e-06
resin2       -1.033      0.543   -1.90  0.065
resin3       -5.800      0.543  -10.69 7.2e-13
resin4       -6.183      0.543  -11.40 1.2e-13
resin.5      -4.800      0.543   -8.85 1.2e-10
```



```

resin6      -5.450      0.543    -10.04   4.1e-12
resin7      -3.692      0.570     -6.47   1.4e-07
resin8      -0.183      0.543     -0.34   0.737
Residual standard error: 0.94 on 37 degrees of freedom
Multiple R-Squared: 0.905, Adjusted R-squared: 0.882
F-statistic: 39.2 on 9 and 37 DF, p-value: 3.00e-16

```

The diagnostics for this model are satisfactory. We see that operator 1 and resin car 1 produce the largest particle size while operator 3 and resin car 4 produce the smallest.

We can construct pairwise confidence intervals for the treatment factor using the Tukey method:

```

> Tukey HSD (aov (psize ~ operator+resin, subset==45))
  Tukey multiple comparisons of means
    95% family-wise confidence level

$operator
      diff      lwr      upr
2-1  -0.2625 -1.0737  0.54872
3-1  -1.7771 -2.6017 -0.95246
3-2  -1.5146 -2.3392 -0.68996

$resin
      diff      lwr      upr
2-1  -1.03333 -2.775204  0.70854
3-1  -5.80000 -7.541871 -4.05813
etc

```

We see that operators 1 and 2 are not significantly different but operator 3 is different from both. There are more differences in the resin cars.

The analysis above is appropriate for the investigation of specific operators and resin cars. These factors are being treated as *fixed effects*. If the operators and resin cars were randomly selected from larger populations of those available, they should be analyzed as *random effects*. This would require a somewhat different analysis not covered here. However, we can at least see from the analysis above that the variation between resin cars is greater than that between operators.

It is important that the observations taken in each cell are genuine replications. If this is not true, then the observations will be correlated and the analysis will need to be adjusted. It is a common scientific practice to repeat measurements and take the average to reduce measurement errors. These repeat measurements are not independent observations. Data where the replicates are correlated can be handled with repeated measures models. For example, in this experiment we would need to take some care to separate the two measurements for each operator and resin car. Some knowledge of the engineering might be necessary to achieve this.

15.4 Larger Factorial Experiments

Suppose we have factors $\alpha, \beta, \gamma, \dots$ at levels $l_\alpha, l_\beta, l_\gamma, \dots$. A *full* factorial experiment has at least one run for each combination of the levels. The number of combinations is $l_\alpha l_\beta l_\gamma \dots$.

which could easily be very large. The biggest model for a full factorial contains all possible interaction terms, which range from second-order, or two-way, as encountered earlier in this chapter, to high-order interactions involving several factors. For this reason, full factorials are rarely executed for more than three or four factors.

There are some advantages to factorial designs. If no interactions are significant, we get several one-way experiments for the price of one. Compare this with doing a sequence of one-way experiments. Also factorial experiments are efficient with experimental resources. It is often better to use replication for investigating another factor instead. For example, instead of doing a two-factor experiment with replication, it is often better to use that replication to investigate another factor.

The analysis of full factorial experiments is an extension of that used for the twoway ANOVA. Typically, there is no replication due to cost concerns so it is necessary to assume that some higher order interactions are zero in order to free up degrees of freedom for testing the lower order effects. Not many phenomena require a precise combination of several factors so this is not unreasonable.

Fractional factorials

Fractional factorials use only a fraction of the number of runs in a full factorial experiment. This is done to save the cost of the full experiment or to make only a few runs because the experimental material is limited. It is often possible to estimate the lower order effects with just a fraction. Consider an experiment with seven factors, each at two levels:

Effect	mean	main	2-way	3-way	4	5	7	7
Number of parameters	1	7	21	35	35	21	7	1

Table 15.1 *Number of parameters in a two-level, seven-factor experiment.*

If we are going to assume that higher order interactions are negligible then we do not really need $2^7=128$ runs to estimate the remaining parameters. We could perform only eight runs and still be able to estimate the seven main effects, though none of the interactions. In this particular example, it is hard to find a design to estimate all the two-way interactions uniquely, without a large number of runs. The construction of good designs is a complex task. For example, see Hamada and Wu (2000) for more on this. A Latin square (see Section 16.2) where all predictors are considered as factors is another example of a fractional factorial.

In fractional factorial experiments, we try to estimate many parameters with as few data points as possible. This means there are often not many degrees of freedom left. We require that σ^2 be small; otherwise there will be little chance of distinguishing significant effects. Fractional factorials are popular in engineering applications where the experiment and materials can be tightly controlled. Fractional factorials are popular in product design because they allow for the screening of a large number of factors. Factors identified in a screening experiment can then be more closely investigated. In the social sciences and medicine, the experimental materials, often human or animal, are much less homogeneous and less controllable, so σ^2 tends to be relatively larger. In such cases, fractional factorials are of no value.

Let's look at an example. Speedometer cables can be noisy because of shrinkage in the plastic casing material. An experiment was conducted to find out what caused shrinkage by screening a large number of factors. The engineers started with 15 different factors: liner outside diameter, liner die, liner material, liner line speed, wire braid type, braiding tension, wire diameter, liner tension, liner temperature, coating material, coating die type, melt temperature, screen pack, cooling method and line speed, labeled a through o. Response is percentage of shrinkage per specimen. There were two levels of each factor. The "+" indicates the high level of a factor and the "-" indicates the low level.

A full factorial would take 2^{15} runs, which is highly impractical, thus a design with only 16 runs was used where the particular runs have been chosen specially so as to estimate the mean and the 15 main effects. We assume that there is no interaction effect of any kind. The data come from Box, Bisgaard, and Fung (1988).

Read in and check the data:

```
> data (speedo)
> speedo
  h d l b j f n a i e m c k g o      y
1 - - + - + + - - + + - + - - + 0.4850
2 + - - - - + + - - + + + + - - 0.5750
.....
16 + + + + + + + + + + + + + + 0.5825
```

Fit and examine a main effects only model:

```
> g <- lm (y ~ ., speedo)
> summary (g)
Residuals:
ALL 16 residuals are 0: no residual degrees of freedom!
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.5825000          NA      NA      NA
h-            -0.0621875          NA      NA      NA
d-            -0.0609375          NA      NA      NA
l-            -0.0271875          NA      NA      NA
b-             0.0559375          NA      NA      NA
j-             0.0009375          NA      NA      NA
f-            -0.0740625          NA      NA      NA
n-            -0.0065625          NA      NA      NA
a-            -0.0678125          NA      NA      NA
i-            -0.0428125          NA      NA      NA
e-            -0.2453125          NA      NA      NA
m-            -0.0278125          NA      NA      NA
c-            -0.0896875          NA      NA      NA
k-            -0.0684375          NA      NA      NA
g-            0.1403125          NA      NA      NA
o-            -0.0059375          NA      NA      NA
Residual standard error: NaN on 0 degrees of freedom
Multiple R-Squared: 1, Adjusted R-squared: NaN
F-statistic: NaN on 15 and 0 DF, p-value: NA
```

There are no degrees of freedom, because there are as many parameters as cases. We cannot do any of the usual tests. It is important to understand the coding here, so look at the *X*-matrix:

```
> model.matrix (g)
(Intercept) h d l b j f n a i e m c k g o
1          1 1 1 0 1 0 0 1 1 0 0 1 0 1 1 0
...etc...
```

We see that “+” is coded as zero and “-” is coded as one. This unnatural ordering is because of the order of “+” and “-” in the ASCII alphabet.

We do not have any degrees of freedom so we can not make the usual F-tests. We need a different method to determine significance. Suppose there were no significant effects and the errors were normally distributed. The estimated effects would then just be linear combinations of the errors and hence normal. We now make a normal quantile plot of the main effects with the idea that outliers represent significant effects:

```
> qqnorm (coef (g) [-1] , pch=names (coef (g) [-1]) )
```

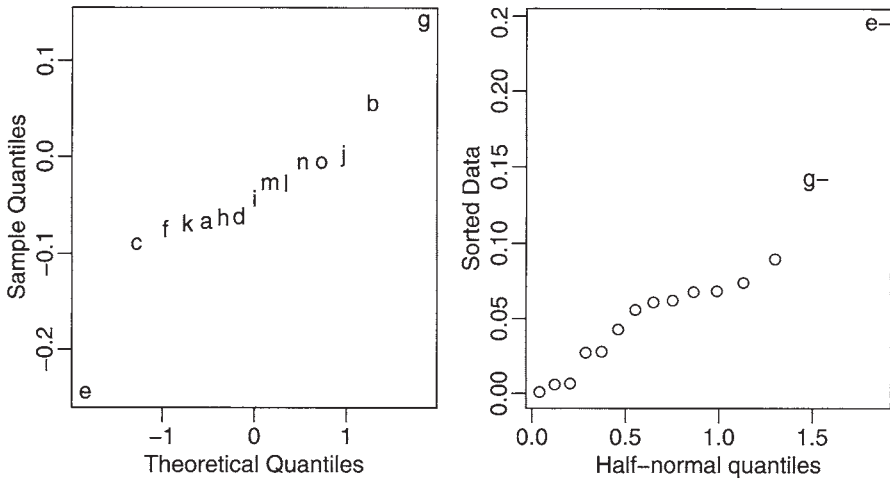


Figure 15.6 *Q-Q plots of effects for speedometer cable analysis.*

See Figure 15.6. Notice that “e” and possibly “g” are extreme. Since the “e” effect is negative, the “+” level of “e” increases the response. Since shrinkage is a bad thing, increasing the response is not good so we would prefer whatever “wire braid” type corresponds to the “-” level of “e”. The same reasoning for “g” leads us to expect that a larger (assuming that is “+”) would decrease shrinkage.

A half-normal plot is better for detecting extreme points:

```
> halfnorm (coef (g) [-1] , labs=names(coef(g) [-1]) )
```

We might now conduct another experiment focusing on the effect of “e” and “g.”

Exercises

1. Analyze warpbreaks data as a two-way ANOVA. Which factors are significant? Now check for a good transformation on the response and see whether the model may be simplified. Now form a six-level factor from all combinations of the wool and tension factors. Which combinations are significantly different?
2. The barley data may be found in the lattice package. Perform a three-way ANOVA with yield as the response. Check the diagnostics—you will find that two points stick out, which correspond to the same variety and site, but for different years. There is reason to suspect that the response values for these cases have been transposed. Investigate the effect of transposing these cases on the analysis.
3. Determine the important factors in the sono dataset where the Intensity is the response and the other variables are predictors.
4. Using the rats data, model the survival time in terms of the poison and treatment. Use the Box-Cox method to determine an appropriate transformation on the response. Which treatments and which poisons are significantly different?
5. The peanut data come from a fractional factorial experiment to investigate factors that affect an industrial process using carbon dioxide to extract oil from peanuts. Determine which factors are important remembering to consider two-way interactions.

